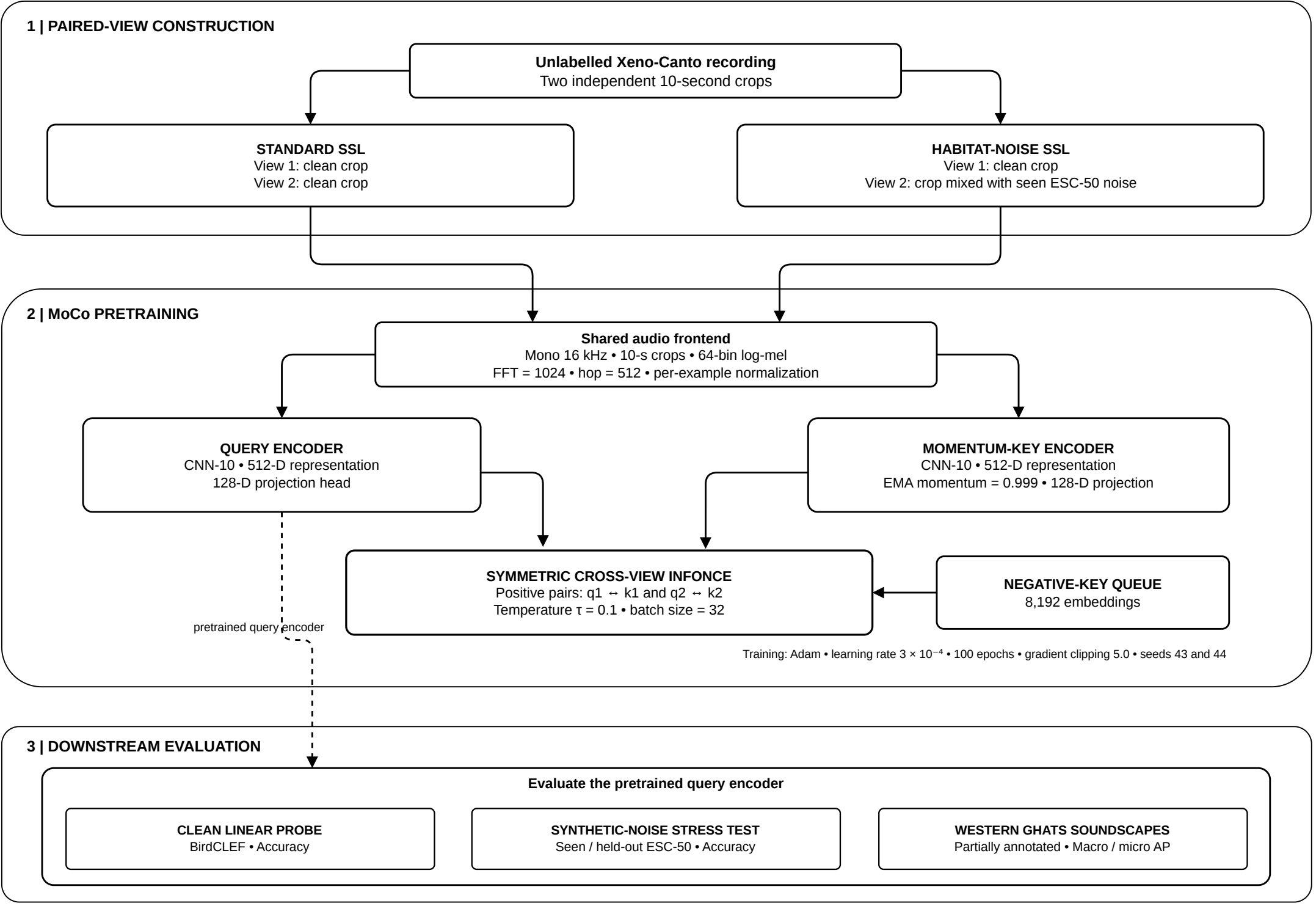
1 | PAIRED-VIEW CONSTRUCTION
Unlabelled Xeno-Canto recording
Two independent 10-second crops
STANDARD SSL
View 1: clean crop
View 2: clean crop
HABITAT-NOISE SSL
View 1: clean crop
View 2: crop mixed with seen ESC-50 noise
2 | MoCo PRETRAINING
Shared audio frontend
Mono 16 kHz • 10-s crops • 64-bin log-mel
FFT = 1024 • hop = 512 • per-example normalization
QUERY ENCODER
CNN-10 • 512-D representation
128-D projection head
MOMENTUM-KEY ENCODER
CNN-10 • 512-D representation
EMA momentum = 0.999 • 128-D projection
SYMMETRIC CROSS-VIEW INFONCE
Positive pairs: q1 ↔ k1 and q2 ↔ k2
Temperature τ = 0.1 • batch size = 32
NEGATIVE-KEY QUEUE
8,192 embeddings
pretrained query encoder
Training: Adam • learning rate 3 × 10⁻⁴ • 100 epochs • gradient clipping 5.0 • seeds 43 and 44
3 | DOWNSTREAM EVALUATION
Evaluate the pretrained query encoder
CLEAN LINEAR PROBE
BirdCLEF • Accuracy
SYNTHETIC-NOISE STRESS TEST
Seen / held-out ESC-50 • Accuracy
WESTERN GHATS SOUNDSCAPES
Partially annotated • Macro / micro AP